

Chapter 6 High-Level and Low-Level Design

6.1 System Overview

In this chapter we explore the comprehensive design of SynapSure a revolutionary medical diagnostic platform that integrates machine learning techniques for improved brain health assessments.

6.1.1 System Design

The SynapSure system architecture incorporates a number of crucial steps that provide an easy move from collecting data to real time analysis.

6.1.1.1 Data Acquisition and Preprocessing

The procedure begins with the collection of various brain scan data. SynapSure accurately classify disorders such as brain tumors, strokes, and hemorrhages by using pre trained models. Data preprocessing is the process of extracting useful information from unprocessed data and improving input.

6.1.1.2 Model Development

The careful development of advanced machine learning models is the foundation of SynapSure. The EfficientNet model a customized convolutional neural network serves as a foundational part which is capable of extracting features required for accurate brain health analysis. SynapSure includes regression models with CNNs providing an additional layer to the systems analytical abilities. This integration allows for more complete evaluation and prediction strengthening the platform's ability to spot abnormalities and measure overall brain health. Through the synergy of CNNs and regression models, SynapSure achieves a harmonious blend of pattern recognition and quantitative analysis. This dual-model approach ensures a robust framework for early detection and precise treatment planning underscoring SynapSure's commitment to advancing neurology diagnostics for improved patient outcomes.

6.1.1.3 Real Time Assessment

SynapSure makes sure users can use the platform smoothly. The interface is designed to help detect problems early which is important for making sure patients get better results in their treatment. This explains that users can quickly and accurately understand what is happening with the brain scans which makes the whole process more effective and helpful.

6.1.2 Data Collection

The SynapSure dataset is gathered with great care in controlled settings which makes sure that it has the best examples for training and testing the machine learning models. This information shows how users behave within the specific limits of the project. The datasets include almost 42,000 brain scans for brain tumor, 12,000 brain scans for brain stroke and around 60,000 brain scans for brain hemorrhage. Having such a diverse and large dataset helps the models learn well and become really good at identifying different brain conditions accurately.

6.1.3 Model Training

During SynapSure model training phase we employ the complicated EfficientNet model with the regression algorithms. For effective utilization of deep learning capabilities in precise and specific brain health diagnosis this strategic choice cannot be overemphasized. The EfficientNet model which is well known for its superior performance in image classification tasks is good at identifying minute trends in a wide range of brain imaging data. The regression techniques help enhance EfficientNet model by providing with continuous readings that help give deeper insight into brain health measures. SynapSure uses this dual approach that combines the strengths of EfficientNets model and regression in order to provide precise and informative assessment of serious conditions such as brain tumor, stroke and hemorrhage. SynapSure aims to optimize the EfficientNet model and regression algorithms by training on a large data set ensuring the system's reliability, adaptability and effectiveness in changing the brain health diagnosis.

6.2 Design Considerations

This section addresses critical issues and considerations essential for SynapSure's development.

6.2.1 Assumptions and Dependencies

6.2.1.1 Assumptions

- The system consider that the collected data is reliable and free from errors.
- SynapSure relies on the proper operation of hardware components such as hardware devices in order to run our website.

6.2.1.2 Dependencies

- SynapSure relies on users being available for brain imaging evaluations to upload the scan.

- SynapSure dependent on internet to upload scans on our website and to generate its report.

6.2.2 General Constraints

SynapSure recognizes limitations that impact its software design such as hardware limits and data standards compliance and performance needs.

6.2.2.1 Key Constraints

- The specs and capabilities of the hardware used for brain imaging restrict system performance.
- Compatibility with defined data standards is critical for scalability and correct operation of system.

6.2.3 Goals and Guidelines

6.2.3.1 Key Objectives

- SynapSure aims for high precision in brain health assessments ensuring accurate classification of conditions.
- The system is designed to be adaptable accommodate future changes in machine learning for effectiveness.
- SynapSure depends on scalability to handle large volumes of brain imaging data and improves diagnostic requirements.

6.2.4 Development Methods

SynapSure uses the Scrum development process which is a flexible and step-by-step way to build and improve the software. With Scrum we can keep adding new features smoothly and adjust things easily if needed. It is like a continuous process of making the software better bit by bit. This makes SynapSure really adaptable and able to meet changing needs. It is like making sure the software is always ready and works well with the latest improvements which creates a system that is both active and good at what it does.

6.3 System Architecture

SynapSure's architecture provides a high level overview of its functionality.

6.3.1 Subsystem Architecture

6.3.1.1 Major Components

- Collects the diverse brain imaging data from the various sources.
- Involves creation and training of machine learning models for the accurate classification.
- Enables users to interact with the platform hence contributing to real time assessments.

6.4 Architectural Strategies

SynapSure employs various strategies to shape its architecture and the decision making processes.

6.4.1 Technology Stack

6.4.1.1 Programming Language

In the backend development of SynapSure we use the Python language which is great because it works well with machine learning libraries and frameworks. For the frontend development we use React a popular technology that helps create a user friendly and interactive interface. This combination of Python for the backend and React for the frontend ensures that SynapSure is equipped with the right technologies to provide a smooth and effective user experience.

6.4.1.2 Database

MongoDB is chosen for its flexibility and scalability in storing and retrieving brain imaging data.

6.4.1.3 Machine Learning Framework

In our methodology we have adopted the EfficientNet model for its superior performance coupled with a regression approach as outlined in the previous chapter. TensorFlow which a powerful and versatile machine learning framework is the engine behind our model development and training. Its robust capabilities play a pivotal role in ensuring the efficiency and effectiveness of our machine learning endeavors.

6.4.2 Scalability and Adaptability

SynapSure's architecture is designed to scale seamlessly accommodating an increasing volume of brain imaging data. The system is built to be adaptable, allowing for the integration of future advancements in machine learning technologies.

6.5 Domain Model/Class Diagram

Following is the model diagram of the project SynapSure.

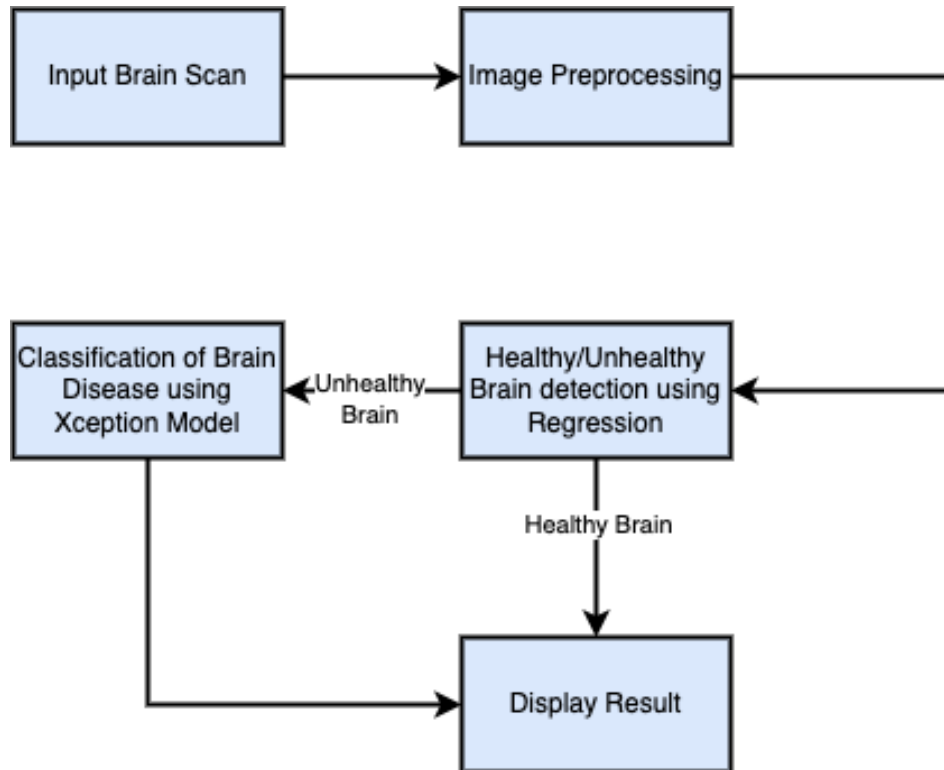


Figure 6.1: Model Diagram of SynapSure

A visual representation illustrating the overall model architecture of SynapSure. This figure shows the steps of model. It starts from inputting the brain scan and then image processing and then regression is done and then it classifies brain disease using EfficientNet model and then it displays the result

The above diagram explains the proposed system. The model begins with the input of brain scans which undergo a image preprocessing stage aimed at enhancing data quality through techniques like normalization and noise reduction etc. The subsequent stages involve feature extraction and selection where relevant patterns and structures in the brain images are identified and isolated for further analysis. The major point in the model is the classification stage, which aims to determine whether the scanned brain is healthy or unhealthy. If the brain is deemed healthy the system displays the result. In cases where unhealthy brain is identified the system further delves into disease classification, employing advanced algorithms to detect specific conditions such as tumors, strokes and hemorrhage. The final step involves displaying the detailed results and suggestions. In summary this model it ensures a comprehensive and effective assessment of brain scans which provides medical professionals with valuable insights into the complex domain of brain health.

6.6 Sequence Diagrams

Following are sequence diagrams that are presented in our web application.

6.6.1 SignUp

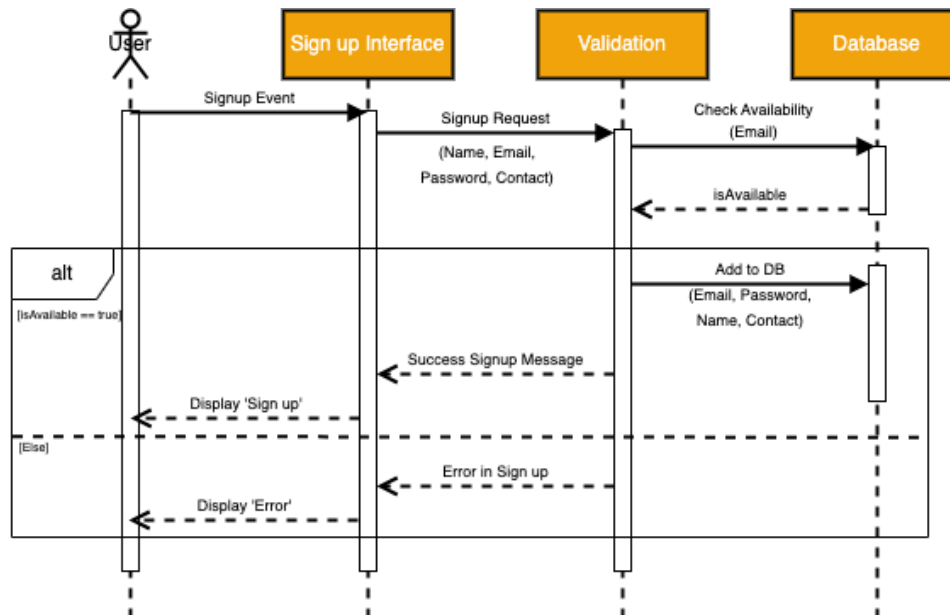


Figure 6.2: SignUp

This figure represents the Signup Sequence Diagram of our system. It shows all the steps starting from sign up event and then request of sign up and it displays the sign up or error

6.6.2 Login

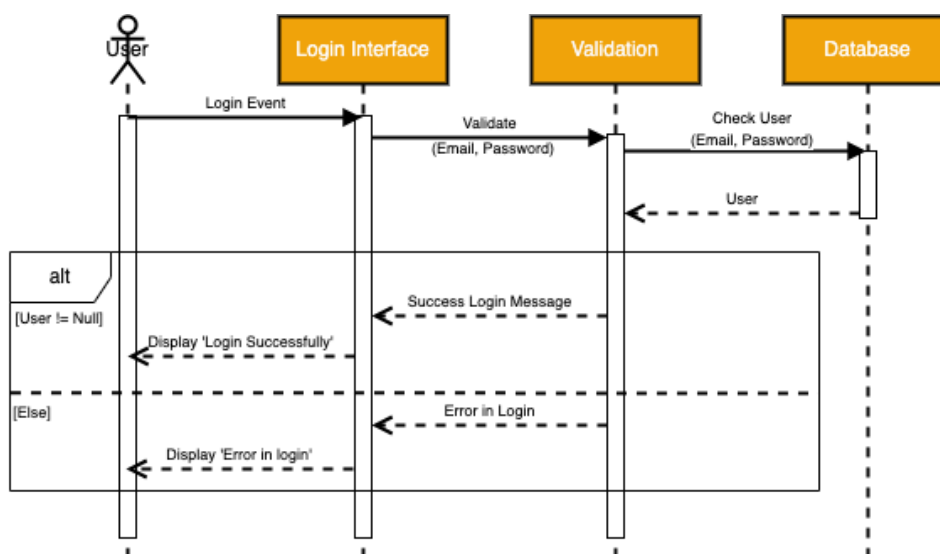


Figure 6.3: Login

This figure represents the Login Sequence Diagram of our system. It starts from the login event and then it checks the user and ends with displaying user login successfully or it displays error

6.6.3 Upload Scan

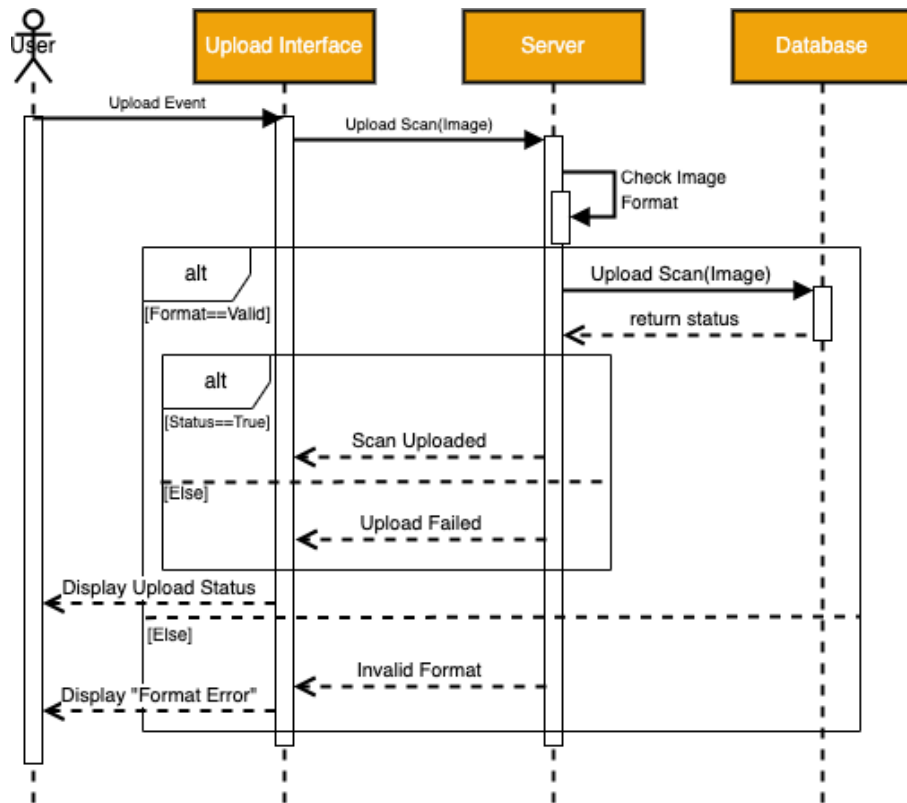


Figure 6.4: Upload Scan

This figure represents the Upload Scan Sequence Diagram of our system. It starts from uploading the event and then upload scans and after processing it shows scan uploaded or upload failed

6.7 Policies and Tactics

This section outlines specific policies and tactics influencing interface and the implementation details of SynapSure.

6.7.1 Software Tools

- SynapSure employs Visual Studio Code and Google Colaboratory for efficient development, testing and deployment.
- Git is used for tracking the changes and collaborating on the code base.

6.7.2 Coding Standards

Coding standards in Python are followed to maintain code readability and consistency. Regular code reviews are conducted to ensure the adherence to the coding guidelines.

6.7.3 Testing Procedures

SynapSure follows a rigorous testing process including unit testing, integration testing and user acceptance testing. Continuous testing ensures the reliability and accuracy of the machine learning models.

6.7.4 Software Maintenance

Regular updates and maintenance procedures are implemented, to keep SynapSure's software up-to-date. Future enhancements including the incorporation of new machine learning algorithms are planned to improve diagnostic capabilities.

6.8 Conclusion

This chapter has provided a detailed exploration of the high level and low level design considerations for SynapSure. From system architecture to design goals and development methodologies the chapter highlights the thought process and decisions, guiding the development of this innovative medical diagnostic platform. The subsequent chapters will delve into the detailed implementation and testing phases hence providing a comprehensive understanding of the SynapSure project.